Tutorial sheet 6 Draft

1. (a) Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}$. Is A diagonalizable? If yes, diagonalize it explicitly, i.e., find P such that ...

(b) Repeat for matrices A, R, B, U, U' written on the board in the image Sep 30-3 (over $\mathbb{R}$ and over $\mathbb{C}$).

(c) For a general 2 by 2 matrix M with entries in $\mathbb{C}$, give a necessary and sufficient criterion in terms of the entries of M for M to be diagonalizable. Does your answer change if $\mathbb{C}$ is replaced by $\mathbb{R}$? What about $\mathbb{Q}$?

2. (From SMMC 2023) Let n be a positive integer. Let A, B and C be three n-dimensional subspaces of $\mathbb{R}^{2n}$ with the property that $A \cap B = B \cap C = C \cap A = \{0\}$. Prove that there exists a basis $\{a_1, a_2, \ldots, a_n\}$ of A, a basis $\{b_1, b_2, \ldots, b_n\}$ of B and a basis $\{c_1, c_2, \ldots, c_n\}$ of C with the property that for each $i \in \{1, 2, \ldots, n\}$, the vectors a_i, b_i and c_i are linearly dependent.

3. The set of invertible $n \times n$ matrices with entries in a field F is denoted $GL_n(F)$ and called the general linear group. Observe that the number of ordered bases of F^n is in bijection with $GL_n(F)$.

(a) Show that $|GL_2(\mathbb{F}_p)| = p(p+1)(p-1)^2$. Find $|GL_n(\mathbb{F}_p)|$. (Optional: Find $|SL_n(\mathbb{F}_p)|$, i.e., count matrices with determinant 1. The key point is that the determinant map from $GL_n(F)$ has uniform fibers. Why?)

(b) How many subspaces of each possible dimension are there in $\mathbb{F}_p^3$?

4. (a) Let P be a linear operator on V. Show that $P^2 = P$ if and only if $P|_{image(P)} =$ identity on $image(P)$. Such a P is called a projection (NOT orthogonal projection: that requires more structure). Is a projection diagonalizable? What is its trace?

(b) Projections are useful because they are closely related to breaking up a vector space as a direct sum. Show that $V = K \oplus W$ if and only if there is a projection P such that $ker(P) = K$ and $image(P) = W$. (Compare Artin's Proposition 4.3.1b.)

Optional but important addendum: Projections are also closely related to direct sum decomposition into *invariant* subspaces. Suppose T is a linear operator on V and K is a T-invariant subspace of V.

Question: Can we write $V = K \oplus W$ with W also T-invariant? In other words, can we arrange T to have a 2×2 block *diagonal* matrix rather than just a block *triangular* matrix? That would break the action of T into two independent parts, which would clearly help in trying to understand T.

Answer: this is possible if and only if there is a projection P with $ker(P) = K$ and $TP = PT$. Prove this.

Further remarks: Operators commuting with T are interesting. E.g., see that the kernel and image of such an operator are T-invariant subspaces. Everything in the addendum becomes transparent when expressed in the language of modules over an appropriate ring.

5. (a) For a vector space V with basis $\{u, v, w\}$, consider the linear operator T defined by $T(u) = v, T(v) = w$ and $T(w) = u$. Write the matrix A of T with respect to the given basis. Take the underlying field to be in turn $\mathbb{R}, \mathbb{C}$ and $\mathbb{F}_p$ where p is a prime number. For each of these cases, decide if T is diagonalizable. For $\mathbb{F}_p$ your answer may be in terms of a condition on the prime p.

(b) A 3×3 matrix B with entries in $\mathbb{C}$ satisfies $tr(B) = tr(B^2) = 0$ and $tr(B^3) = 3d$ for some given *nonzero* scalar d in $\mathbb{C}$. Find $det(B)$. Show that all such matrices are similar to each other. (Square matrices A and B are called similar or conjugate if $B = P^{-1}AP$ for some invertible matrix P.)

6. Recall that to analyze a linear operator T on a vector space V, it may help to decompose V as a direct sum of T-invariant subspaces. (a) Solve problem 5.9 in chapter 4. (b) Then solve problem 5.10.

7. (a) For a diagonalizable operator T on V, consider the eigenspace decomposition $V = \oplus_\lambda V_\lambda$. Show that any T-invariant subspace U of V is of the form $\oplus_\lambda W^\lambda$, where W^λ is a subspace of V_λ. There is something to prove here! Is $T|_U$ diagonalizable?

(b) Suppose T and S are both diagonalizable operators on a finite dimensional vector space V. Show that if $TS = ST$ then there is a single basis of V with respect to which both S and T have diagonal matrices. The key is to consider an entire eigenspace of T from the point of view of S. (Quick: is the converse true?) Now show that any family (even infinite) of mutually commuting operators on V can be simultaneously diagonalized.

8. The Cayley-Hamilton theorem states that any linear operator T on a finite dimensional vector space V over any field $\mathbb{F}$ satisfies its own characteristic polynomial $p_T(x)$, i.e., $p_T(T) =$ the 0 operator.

(a) Prove the theorem for diagonal(izable) T. (When $\mathbb{F} = \mathbb{C}$, this can be combined with continuity to give the result for any T. See Artin 5.2 for this interesting proof.)

(b) Prove the theorem for triangular(izable) T. Note that this proves the theorem for all T if $\mathbb{F} = \mathbb{C}$. Why?

9. Prove the Cayley-Hamilton theorem for arbitrary T using the outline below. I learned this proof from Vighnesh. It may help to review 8(b).

(a) Solve problem 5.6 in chapter 4. Show that $p_T(T)(v_0) = 0$ vector. Deduce quickly that $p_T(T) = 0$ operator.

(b) Let T be any operator on V. Start with any nonzero vector v_0 and generate a T-invariant subspace U_1 as in part (a) with basis $v_0, \ldots, v_n$. If $U_1 = V$ we are done. Otherwise take a vector w_0 outside U_1. Keep applying T as long as you get vectors linearly independent of ALL previously chosen basis vectors including those in U_1. Name the resulting T-invariant span of $v_0, \ldots, v_n, w_0, \ldots, w_m$ as U_2. What is the characteristic polynomial of $T|_{U_2}$? Show that $T|_{U_2}$ satisfies its own characteristic polynomial. Continue until one exhausts V. You need not make careful notation for U_3 onwards but clearly explain the general pattern in words.

(c) Is the result in problem 5(b) true if we replace every $\mathbb{C}$ by $\mathbb{R}$? (So similarity is also required to be over $\mathbb{R}$.)

10. Suppose $p_T(x) = \prod_i (x - \lambda_i)^{m_i}$ with λ_i distinct. Are λ_i and m_i unique?! Show that for each i, we have

$$\dim \ker(T - \lambda_i I) \leq m_i \qquad \textit{and} \qquad \dim \ker\big((T - \lambda_i I)^\infty\big) = m_i.$$

Show that for distinct i, $\ker\big((T - \lambda_i I)^\infty\big)$ are independent subspaces. Deduce that $V = \bigoplus_i \ker\big((T - \lambda_i I)^{m_i}\big)$. For the direct sum, in class I suggested using gcd's of polynomials. But a different tool we developed helps!

Remarks: (1) $\ker(T - \lambda_i I)$ is just the λ_i-eigenspace, whose dimension is called the *geometric multiplicity*. So the first claim says that geometric multiplicity $\leq$ algebraic multiplicity m_i, and we get equality if we replace the eigenspace by the *generalized eigenspace* $\ker\big((T - \lambda_i I)^\infty\big)$. Is the claim true if $p_T(x)$ doesn't factor fully?!

(2) Note that each direct summand $\ker\big((T - \lambda_i I)^{m_i}\big)$ is a T-invariant subspace, on which $N = T - \lambda_i I$ is a nilpotent operator. This direct sum then feeds into the last and arguably *the* key step of the Jordan form result: *a space with a nilpotent operator N decomposes into a direct sum of subspaces, each of which has a basis on which N just acts by a shift.* Putting everything together, we first decompose V as a macro direct sum of generalized eigenspaces, each of which breaks up further into a micro direct sum of Jordan blocks. An elementary proof of the Jordan form, following Evan Chen's napkin, will be done in a later (optional) class.

(3) Optional food for thought: the direct sum claim does not even make sense if $p_T(x)$ doesn't factor fully. Can anything be salvaged? Yes, because $p_T(x)$ still factors as a product of "prime" polynomials (just like integer factorization – study of polynomial factorization is now unavoidable). Can you guess what the direct sum result should be in this setting?

Next, the Jordan refinement is no longer available, but one can still get to special matrices, namely like the one in Artin problem 4.5.6 in 9(a) above. Echo from the past: Optional problem 9(c) from Tutorial sheet 3

about integer matrices under row/column equivalence is highly relevant here. The relevance is not obvious and is probably best developed in its natural setting, namely the study of modules over the polynomial ring $F[x]$ (also alluded to in the remark after problem 4). However, figuring out canonical representatives for the equivalence classes is the decisive step, and this works the same way over $\mathbb{Z}$ and $F[x]$, by using long division.

11. Matrix algebra on steroids (true meaning of finite direct sums).

Work over some fixed field F. Let $V_1, V_2, \ldots, V_c$ be vector spaces and let V be their direct sum. Let $W_1, W_2, \ldots, W_r$ be vector spaces and let W be their direct sum. Let U be a vector space.

(a) Show that the set of linear maps from U to W is in one-to-one correspondence with r-tuples $(f_1, \ldots, f_r)$ where f_i is a linear map from U to W_i, as i goes from 1 to r. Should the r-tuple be symbolically denoted by a row or a column?

(b) Formulate an analogous statement about the set of linear maps from V to U and prove it.

(c) Show that the set of linear maps from V to W is in one-to-one correspondence with the set of $r \times c$ matrices, where the (i, j)-th entry is a suitable linear map that you should clearly specify.

(d) Show how our earlier encoding of linear maps by matrices of scalars (via choosing bases) is just a special case of this more general situation.

(e) Going the whole hog: Using the identification from part (c) (of a linear map from V to W as a matrix of smaller linear maps), translate the operations of addition of linear maps in $\mathrm{Hom}(V, W)$ and composition of linear maps (given $X \xrightarrow{S} V \xrightarrow{T} W$ with X also a direct sum) into operations on these matrices. Show how this yields the standard definitions of matrix addition and matrix multiplication.

Remark: Later on, you will see that these results go through exactly the same way when we replace the ground field F by any ring and vector spaces by modules in parts (a)-(c),(e) and by free modules in part (d).